

Xinyue (Cindy) Wang

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EDUCATION

Carnegie Mellon University

Major in Computer Science and Information Systems

Relevant Coursework

Computer Systems, Imperative Computation, Database Development, Artificial Intelligence, Foundations of Software Engineering, Application Development, Functional Programming

Activities/Interests: CMU Club Volleyball, Dance, Chinese Student Association, Baking, Photography

Pittsburgh, PA

Expected May 2028

EXPERIENCE

Software Engineer Intern

Jun 2026 – Present

LeapBound AI

San Jose, CA

- Independently designed and shipped two production, full-stack AI brand-intelligence engines targeted at small-to-medium Chinese companies entering the US market: an **SEO** audit surfacing technical fixes and content-credibility gaps, and a **GEO** audit scoring **AI citation rate** and **share-of-voice** against competitors, live at leapbound.ai.
- Built both tools end-to-end on Cloudflare and Supabase, owning the Python audit pipeline, FastAPI endpoints, **database schema/RPCs**, and Next.js frontend, including an **LLM-guided crawler** that curates each sitemap to relevant pages.
- Engineered the SEO audit's crawl-to-report pipeline: deep site-crawls and Google PageSpeed data feed a **deterministic** technical analyzer (duplicate titles, meta descriptions, schema) plus a credibility check that flags unsupported marketing claims, output as a prioritized 30-day plan with hour estimates.
- Implemented the GEO evaluation harness: generates market-aware buyer questions, extracts and verifies brand citations in LLM responses (blocking **hallucinated** competitor mentions), and computes a composite visibility score from citation rate, content readiness, and authority signals grounded in Wikipedia, Wikidata, and Common Crawl, surfacing false claims AI assistants make about client brands.

Software Engineer Intern

Apr 2026 – Present

Howbe Inc.

San Jose, CA

- Built and shipped core features for WhoBe, a Flutter-based personal CRM app, spanning backend API design, data-layer consistency, and reactive frontend state management.
- Hardened the **read/write consistency** of a Google Sheets-backed data layer (dynamic schema-aware reads, redesigned caching), then built a **bulk aggregation API** that replaced $O(n)$ per-contact requests with a **single round-trip**, powering live unread indicators and custom sort ordering.
- Developed a **reactive client-side state layer** (coalesced refresh, persisted read state) driving live Flutter UI updates without manual reloads, backed by **~30 new unit and integration tests** across pytest and the Flutter suite.

Researcher - Cyber Defense Automation

Sept 2024 – Present

Carnegie Mellon School of Computer Science, Department of Defense

Pittsburgh, PA

- Built the extraction pipeline for **CyberKG**, a cyber-defense knowledge graph enabling AI-generated incident-response playbooks across **183 MITRE D3FEND** techniques; co-authored a paper on this work submitted to the **2026 IEEE Conference on Communications and Network Security (CNS)**.
- Designed an LLM-based extraction methodology (**Chain-of-Thought reasoning**, ReAct-style decomposition), constraining the model to each technique's official text and requiring **evidence-grounded** justification per predicate to reduce **hallucination**.
- Achieved **88.1% precondition** and **89.8% postcondition** extraction accuracy across all 183 techniques, validated using an **LLM cross validation** and expert manual review.

PROJECTS

Dynamic Memory Allocator | C, malloc/free, Segregated Free Lists, Heap Utilization

- Built a dynamic memory allocator (malloc, free, realloc, calloc) using **segregated free lists**, with a custom **heap consistency checker** validating block alignment, header/footer integrity, and coalescing correctness across a full 64-bit address space.
- Eliminated block footers and minimized per-block overhead to cut internal fragmentation, achieving **>75% heap utilization** and **>10,000 kops/sec** throughput.

Creamery Management Platform | Ruby on Rails, React, SQLite, CanCanCan, MiniTest, Cucumber

- Built a full-stack employee, store, and payroll management platform (**8 relational models**, **~15 controllers**, a JSON REST API, two React dashboards), delivered test-driven at **100% code coverage**.
- Engineered session-based authentication (has_secure_password) and **role-based authorization** with CanCanCan, enforcing granular admin/manager/employee permissions at both the controller and view layers.
- Developed a versioned JSON REST API (**jsonapi-serializer**) and two React front-end views, including a live time-clock dashboard that drives payroll calculation through real-time clock-in/clock-out state transitions and asynchronous data fetching.

SKILLS

Languages: Python, C, Java, SQL, JavaScript, HTML/CSS, R, Ruby (Rails), Assembly

Tools: Git, GitHub, CI/CD, Docker, FastAPI, RESTful APIs, Flutter, React, Redis, Supabase, Cloudflare